

**LEARNING-ENHANCED ROBUST
ADAPTIVE CONTROL OF NETWORKED
ROBOTIC SYSTEMS UNDER SENSOR AND
COMMUNICATION UNCERTAINTY**

Research Paper

Abulfat Farajov

MSc in Control in Technical Systems

Baku Higher Oil School

Information Technology Department

Baku, Azerbaijan

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Abstract

Networked robotic systems can experience multiple sources of uncertainty, including imperfect sensor measurements, communication delay, packet loss, uncertain system dynamics, and external disturbances. This study investigates whether learning-based uncertainty estimation can provide additional benefits when integrated with a robust adaptive control framework for a two-degree-of-freedom robotic manipulator. An adaptive sliding-mode controller is used as the baseline control mechanism, while a radial basis function neural network (RBFNN) is introduced to estimate lumped system uncertainty. The proposed framework is evaluated using MATLAB-based numerical simulations under sensor noise, sensor degradation, communication delay, packet loss, changing robot dynamics, and external disturbances. Under the combined networked-uncertainty scenario, ten independent simulation runs showed mean tracking RMSE reductions of 5.35% for the first joint and 18.17% for the second joint when the RBFNN was integrated with the adaptive sliding-mode controller. However, the learning component did not provide additional tracking improvement when sensor degradation was included, and no improvement was observed in the changing-dynamics and external-disturbance experiment. These results indicate that the effectiveness of learning-based uncertainty estimation is dependent on the type and combination of uncertainty affecting the closed-loop system rather than being uniformly beneficial across all operating conditions.

Keywords

Networked robotics; robust adaptive control; adaptive sliding-mode control; machine learning; RBF neural network; uncertainty estimation; sensor uncertainty; communication delay; packet loss

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Introduction

1.1 Background and Motivation

Robotic systems increasingly need to function in environments where accurate system models and reliable measurements and communication cannot always be obtained. Uncertainties in practical robotic manipulators can occur due to unmodeled nonlinear dynamics, parameter variations, friction, external disturbances, measurement deviation, and component fault. Recent studies on learning-based control have thus pursued the idea of using neural-network models to account for unmodeled dynamics and disturbances, to enhance trajectory-tracking performance [1]. But other studies have considered sensor faults without making any assumption about the exact bounds of the faults using an adaptive observer and a fault-tolerant controller [2].

In networked robotic systems, the communication-related effects may also impact the control loop. Discrepancies between the available information about the system and the actual state of the robot can arise due to time varying latencies, packet transmission limitations, and other imperfections on the network. For instance, recent research on networked robotic manipulators has explicitly taken into account time-varying communication delays along with external disturbance and system nonlinearities [3]. In general, networked control research has shown that channel delay, communication noise, and packet loss can have measurable consequences on tracking performance [4].

Learning-based approaches are also explored as a way to directly represent uncertainty in the control framework. Han et al. introduced a strong learning based control framework, where a learned model is used to estimate the uncertainty of nonlinear dynamical systems and the uncertainty information related to the dynamical systems is fed into the closed-loop control design [5]. In more recent studies, learning-based robust control has been applied to robotic manipulators where multiple sources of model uncertainty, such as deviations in measurements and unmodeled mechanical effects, have been considered and tested in experiments on a two-degree-of-freedom robotic platform [6].

The developments show that learning, robust control, adaptive control, state estimation and networked control are proven methods to handle various types of uncertainty in robotic systems. When multiple uncertainty sources are present in the problem, however, the practical control problem is more complicated. However, a robotic controller may have to work when the sensor measurements are inaccurate, the communication is delayed or lost, the system dynamics are not fully understood, and there is at the same time external disturbances. This encourages research on architectures for control where the uncertainty information can be estimated and used in the control process as a whole, not independently at each uncertainty source.

The present study thus explores a networked robotic control problem in the presence of sensor and communication uncertainty, uncertain system dynamics and external disturbances. The direction of research is to investigate if learning-based uncertainty estimation can supply

a useful signal to a robust adaptive controller whilst maintaining the robustness and adaptation properties of the controller. This will be tested experimentally by controlled comparison with appropriate baseline controllers and not assumed a priori.

1.2 Problem Statement

In networked robotic systems, there is uncertainty not only in the actual system, but also in the information available to the controller, with which satisfactory control performance being maintained. Uncertainty in robotic manipulators can be caused by failure of sensors or measurement error, unmodelled dynamics, parameter changes, friction, and external disturbances [1, 6]. These effects can decrease the accuracy of the state estimates and result in a decrease in the accuracy of the trajectory-tracking performance.

Problems are more difficult if the robot is controlled via a communication network. Due to time-varying communication delays, the controller might use outdated state information, and packet loss may make it impossible to receive measurements or/or control-related information at the correct time [3, 4]. Some previous research has included the role of network-induced delay and disturbance as an important factor that impacts the attainable operating region and control performance of robotic manipulators [3]. In the same fashion, uncertainty due to the faults of sensors has been dealt with by adaptive observers and fault-tolerant control techniques [1].

Another method to deal with uncertainty is to learn the unmodelled or unseen parts of a robotic system, which is called learning-based control. There have been recent efforts to incorporate learning-based uncertainty representations into robust control problems for nonlinear systems [5]. Another research area is neural-network based robust control of robot manipulators with model uncertainty and measurement deviation [6]. But these methods alone do not provide a way to represent and integrate uncertainty from sensing and communication as if they are both occurring at the same time in an adaptive robust control mechanism.

The problem considered in this research is that of controlling a robotic manipulator in the presence of sensor uncertainty, communication induced uncertainty, uncertain system dynamics, and external disturbances possibly occurring simultaneously. The key technical challenge to solve is whether online learning-based uncertainty estimation can give a useful and informative signal to a robust adaptive controller to ensure acceptable tracking and robustness properties under these combined conditions.

The research doesn't suggest that the learning-based controller will give better performance than the conventional ones. Instead, its performance should be assessed experimentally under specified uncertainty conditions (which may be either isolated or combined uncertainty) with comparison to suitable baseline controllers.

1.3 Research Gap

There have been several individual sources of uncertainty studied in the context of robotic and networked control systems. Adaptive observers and fault-tolerant control methods have

been studied for sensor faults and measurement related uncertainty [2]. The effects of time-varying communication delays and disturbances on the operational behaviour of robotic manipulators have been studied separately [3]. Robust control for uncertain nonlinear systems has been also developed, using learned uncertainty information in the closed-loop controller [5]. Recently, the experimental testing of learning-based robust control has been studied for modular robot manipulators with model and measurement uncertainties [6].

The above research examples show that learning based uncertainty modelling, robust control, sensor fault handling and networked control are well-defined research areas. The literature that was taken into account for this study, however, suggests that they tend to be studied individually or with a small number of uncertainty sources. In a networked-control study, for instance, attention is mainly paid to the communication-impairing aspects (e.g., latency variations and disturbances) while a sensor-focused study focuses primarily on the faults in the measurements.

Another drawback is that the role of learning is not always part of the controller and is not always provided as an explicit estimate of the closed-loop system uncertainty. It is noted that, in learning-based robust control, uncertainty information learned can be embedded in the control law, while the uncertainty induced by the communication in the networked robotic systems may not be treated as another part of the same uncertainty-estimation mechanism [5]. On the other hand, it is possible to explicitly model communication delays and disturbances in the networked robotic control studies but without learning an estimate of uncertainty into the robustness or the adaptation mechanism [3].

One such research gap is the lack of research on a unified robotic control architecture that incorporates learning-based uncertainty estimation to support a true robust adaptive controller with sensor uncertainty and communication-induced uncertainty. We will study this gap by running controlled experiments with the ability to add uncertainty sources one by one and evaluate the learning-based uncertainty estimation component by ablation and baseline comparison.

1.4 Research Questions

The following questions have been used to inform this research:

RQ1. Does uncertainty estimation through learning boost the robustness of a robotic manipulator in uncertainty of sensors and communication?

RQ2. What are the performance characteristics of the proposed learning enhanced robust adaptive controller when the system dynamics change, external disturbances are present, communication delay occurs, sensors are not reliable, and there is packet loss?

RQ3. What happens when several uncertainty sources are at the same time?

RQ4. How much does the uncertainty estimation based on the learning improve the control performance compared to robust and adaptive controllers that do not have the learning component?

RQ5. What would be the impact of the proposed approach on tracking accuracy, control effort, disturbance rejection, degradation of robustness and computational cost?

1.5 Contributions

1. The following are some of the desired outcomes of this research:
2. Learning based uncertainty estimation and a strong adaptive controller of a networked robotic manipulator with sensor and communication uncertainty.
3. An uncertainty estimation mechanism to inform the control framework of uncertainty conditions relevant to the control, but not to supplant the robustness of the underlying adaptive control mechanism.
4. Systematic experimental testing of sensor noise and degradation, communication delay, packet loss, external disturbances, and changing system dynamics in isolation and in combination.
5. A comparative assessment with relevant baseline controllers to assess the impact of the learning component on tracking performance, robustness, control effort and computational burden.
6. An ablation analysis to determine if learning-based uncertainty estimation can make a measurable contribution to control performance.

Such contributions are tentative, and will be adjusted once the literature review and final methodology have been decided.

1.6 Paper Organization

The rest of this paper is organized as follows. Recent advances in robust and adaptive control, learning-based control, sensor uncertainty and state estimation, and networked robotic control are summarized in Section 2. The robotic system model, sensor model and communication model, uncertainty and disturbance modeling and the control goal are presented in Section 3. The proposed methodology, such as state estimation, learning-based uncertainty estimation, and robust adaptive control, is described in Section 4. Section 5 outlines the experimental methodology, baseline controllers, uncertainty scenarios and evaluation metrics. The experimental results are given in section 6. The results, comparisons with previous approaches, limits, and implications are discussed in section 7. Lastly, the conclusions and future work are presented in Section 8.

Related Work / Literature Review

2.1 Robust and Adaptive Control of Robotic Systems

Robust and adaptive control methods have been extensively investigated for robotic manipulators operating in the presence of model uncertainty, external disturbances, nonlinear dynamics, and measurement imperfections. A common objective of these methods is to maintain tracking performance without requiring exact knowledge of all system parameters or disturbance characteristics.

Adaptive control approaches typically introduce online parameter-estimation or adaptation mechanisms to compensate for uncertain system properties. For example, recent work has developed adaptive controllers for robotic manipulators subject to internal uncertainties and external disturbances, demonstrating trajectory-tracking control without assuming that these effects are completely known [7], [8]. Other approaches combine adaptive laws with disturbance observers to estimate unknown disturbances and uncertainty bounds during operation [9].

Robust sliding-mode-based methods remain particularly common because of their ability to tolerate bounded uncertainties and disturbances. Sun et al. combined a nonlinear disturbance observer with an adaptive predefined-time sliding-mode controller, using the observer to estimate disturbances and an adaptive law to estimate the unknown upper bound of system uncertainty [9]. Similarly, Zhang et al. proposed a robust adaptive fast terminal sliding-mode controller for a two-link robotic manipulator with uncertainties and external forces [10]. These approaches demonstrate that adaptive robustness can be maintained without precise prior knowledge of uncertainty magnitudes.

Measurement uncertainty has also been incorporated directly into robust adaptive control formulations. Wang et al. considered robotic systems with measurement noise and developed an adaptive robust prescribed-performance controller in which fuzzy-set methods were used to quantify uncertainty [11]. Other recent research has addressed uncertain manipulator dynamics through observer-based adaptive and model-free control structures, combining online estimation with robust trajectory-tracking mechanisms [12].

Although these methods provide strong mechanisms for handling physical uncertainties and disturbances, their uncertainty treatment is generally based on analytical observers, predefined uncertainty representations, adaptive bounds, or fuzzy approximations. The controller therefore does not necessarily learn the uncertainty characteristics directly from data. This distinction is important for the present study, where learning-based uncertainty estimation is intended to complement, rather than replace, the underlying robust adaptive controller.

2.2 Learning-Based Control

Learning-based control is an important method for robotic systems operating with uncertain or poorly modelled dynamics. Learning methods can also estimate unknown dynamics, disturbances or control policies from data of a system, rather than relying solely on an exact analytical model. Some research has recently been developed to compensate for unmodelled dynamics and external disturbances through neural-network models used with traditional control elements [1] and to model uncertain manipulator dynamics with adaptive neural networks to enhance tracking performance [13], [14].

There is another approach to learning, which is reinforcement learning, that is well established. The RL-based adaptive control has been studied for uncertain mechanical systems that are modeled by Euler–Lagrange equations and the uncertain dynamics and

unknown perturbations are compensated by the learning process [15]. Recently, event-triggered learning control was proposed for uncertain dynamics for robotic manipulators, which integrates adaptive dynamic programming and an event-triggered mechanism to minimize communication and computation resources requirements [16].

A similar approach is the approach of robust learning-based control, where the learning model is not used to compute the control input, but instead used to estimate the uncertainty, which is then passed into a stabilizing controller. Han et al. created a data-driven robust control framework in which the deep stochastic Koopman model learns uncertainty in the nonlinear dynamics and the uncertainty representation learned from the deep stochastic Koopman model is utilized within the closed-loop controller [5]. This approach was tested in simulation and on a real soft robotic arm.

A recent study also shows growing coupling between neural learning and adaptive control. For instance, a 2024 neuroadaptive controller for robotic manipulators is developed to deal with the dynamic-model uncertainty and a disturbance observer to handle non-parametric disturbances, and both numerical and experimental validation are provided [17]. Adaptive and fault-tolerant methods using neural networks have been used for manipulators where parameters are unknown and actuators are faulty [18].

In robotic control, learning can serve a number of purposes: to approximate unknown dynamics, to estimate disturbances, to learn a control policy, or to represent uncertainty in use by another controller. The latter is especially significant for the present research. The goal is not to abandon good adaptive control for a black-box learned policy, but to determine if a policy learned estimate of uncertainty can be useful to a true robust adaptive controller with uncertainty being present in both sensor and communication.

2.3 Sensor Uncertainty and State Estimation

Feedback control relies on accurate state information, however, robotic systems frequently have noisy, incomplete, or degraded measurements. There are many sources of uncertainty in sensors, such as measurement noise, sensor degradation, sensor fault, limited availability of measurements, and uncertainty of the underlying system model. For this reason, state estimation is usually employed to estimate the values of a system's variables that are not measured directly or to minimize the impact of inaccurate measurements [2, 19].

The Kalman-filter-based state estimation methods are still popular when measurement and process noise exists. For robotic systems, adaptive variants can consider changes in models and noise characteristics. Zhang et al. proposed an adaptive extended Kalman filter for the control of a space manipulator based on noisy position measurement while taking into account explicitly process noise, measurement noise and sensor limitations [19]. A sliding-mode state estimation combined with a fault-tolerant control algorithm was implemented, and tested by simulation, both numerically and by Monte Carlo simulation.

If the system model and/or the uncertainty characteristics are not exactly known, robust and adaptive filtering techniques can be used as an alternative. Several adaptive Smooth Variable

Structure Filter – Kalman Filter (SVS–FK) approaches for state estimation under faults and uncertain system conditions were compared by Goodman et al. [20]. Their findings show the differences between traditional stochastic estimation and robust estimation methods which will be resistant to changes in the assumed system model or to model inaccuracy.

Sensor faults are more serious types of measurement uncertainty since the measured signal may not be reflective of the physical condition of the system. Ebrahimi et al. proposed an adaptive observer for robotic manipulators which determined and estimated sensor faults without knowing any bounds on the fault and its derivative [2]. The estimated information is then fed into an observer-based fault-tolerant controller, highlighting the possibility of integrating state estimation with control in order to achieve good performance in the presence of degraded measurement data.

More recent learning-based methods show another direction to which state estimation can be learned from data. Specifically, a study done at IROS 2025 suggested using a neural network based state estimator to also estimate the state uncertainty for multi-robot coordination [21]. In that scheme, the "uncertainty" is incorporated into the decision for control by adjusting the contribution of uncertain state information.

Therefore, for the present research, the aim of state estimation is essentially considered as a transitional step between the sensing and control. The ultimate goal is not just to obtain a filtered estimate of the state, but to explore if it is possible to learn information about the reliability or uncertainty of that estimate and then feed it into the robust adaptive controller. This establishes a direct link between the uncertainty estimation of the sensors and the proposed learning-enhanced control architecture.

2.4 Networked Robotic Control

The introduction of communication-related uncertainty in the feedback loop is caused by networked robotic systems. Delay changes may lead to a delay in the controller reacting to measurements, and packet loss may lead to loss of state or control information. These effects can directly impact the tracking performance and under sufficiently adverse conditions, closed-loop stability [3], [4].

These effects have recently been analysed, in particular, in robotic manipulators. Huyn et al. studied networked robotic manipulators with time varying communication delays and external disturbances, and proposed a technique to characterize the minimal operation regions under such conditions [3]. They employed a non-linear robotic arm model, and tested their analysis by simulation.

Other networked-control research has been on the theoretical impact of packet loss and delay on control performance. Lu et al. studied the performance bounds of networked control systems when channel delay and packet dropouts are introduced in the system [4]. Liu et al. later expanded the consideration of the packet dropouts to include the input delay, and presented an estimation and control system for the UDP communication scenario [22].

Learning methods are also starting to be used in networked control. In [23] a 2025 study was conducted on an H^∞ tracking problem with network delays, packet loss and external disturbances occurring concurrently. A predictor was used to account for the influence of the delayed and missing state information. A more recent study explored a networked rehabilitation robot in the presence of persistent uncertainty and quantization of the inputs to the controller, with a real-time neural-network adaptation in the controller structure [24].

Event triggered transmission and quantization have also been taken into account as means of reducing communication requirements in the communication-aware approaches. Xiang and Liu proposed an event-triggered, quantized networked-control architecture explicitly considering communication delay and packet dropouts [25]. The studies show that communication constraints can be directly addressed in the control design and not just as external perturbations.

It is important to note that communication uncertainty is not looked upon as an individual networking problem, but rather as a part of closed-loop uncertainty for the present research. The effects of these variables and unknowns will be measured in combination with sensor uncertainty and uncertain robot dynamics: communication delay and packet loss.

2.5 Research Gap

The literature reviewed indicates that research has already been conducted on the various aspects of a research direction proposed here. The robust and adaptive controllers [7]–[12] address the uncertainties of robot dynamics and disturbances, the learning-based controllers estimate or compensate for the uncertainties of the robot dynamics and disturbances [1]–[5, 13]–[18], and the networked-control ones address the communication delay, packet loss and related limitations [3, 4, 22]–[25].

When several uncertainty sources are present, the real question is then how to combine the various techniques. Specifically, the presented studies offer very weak evidence of a single robotic architecture and control that explicitly incorporates learned uncertainty information from the imperfect sensing and network-induced uncertainty and then exploits it to steer a robust adaptive controller. This should be considered as a candidate research gap, rather than as a novel claim.

This candidate gap will be examined through the proposed research by splitting up the learning and control functions. The learning controller will provide uncertainty information relevant to the system of interest from the information in the communication available and the robust adaptive controller will continue to perform closed-loop control. The proposed architecture will, then, be tested in isolated and combined uncertainty scenarios. This design also allows for ablation studies to be performed and evaluate if the learned uncertainty information can contribute measurably to the underlying robust and adaptive controller.

The distinction is key since its combination with networked control and uncertainty handling has already been demonstrated in recent research [23] and [24]. Thus, the ultimate contribution can't just be a “mash-up of machine learning with networked robust control.”

The contribution should rather be based on the specific uncertainty-estimation mechanism, how this mechanism is integrated into the controller and the experimental evidence gained under simultaneous uncertainty conditions.

System Model and Problem Formulation

3.1 Robotic System Dynamics

The robotic plant is modelled as a nonlinear rigid-body manipulator with n degrees of freedom. The standard joint-space dynamics are represented as

Where q denotes the vector of joint positions, $\dot{q}$ the joint velocities, and the $\ddot{q}$ the joint accelerations. $M(q)$ represents the inertia matrix, $C(q, \dot{q})\dot{q}$ represents the Coriolis and centrifugal terms, $G(q)$ represents the gravitational forces, and $F(\dot{q})$ represents friction effects. The vector τ denotes the applied control torque, while $d(t)$ represents external disturbances and unmodelled effects.

The controller is not assumed to have exact knowledge of all physical parameters of the robotic system. Consequently, differences between the nominal model used for control and the actual robot dynamics are treated as system uncertainty. This uncertainty may arise from parameter variations, unmodelled dynamics, friction, and other physical effects.

The primary control objective is to make the actual joint trajectory $q(t)$ track a desired reference trajectory $q_d(t)$. The position tracking error is defined as

$$e(t) = q(t) - q_d(t)$$

and the corresponding velocity error is

$$\dot{e}(t) = \dot{q}(t) - \dot{q}_d(t)$$

The objective is to maintain bounded tracking error and, under appropriate operating conditions, drive the tracking error towards zero despite system uncertainty, external disturbances, imperfect measurements, and communication-related effects.

A 2-DOF robotic manipulator is considered as a candidate experimental platform because it provides nonlinear manipulator dynamics while remaining computationally manageable for simulation and controlled experimentation. The final robotic platform and its parameter values will be determined during the experimental-design stage.

3.2 Sensor Model

The robotic controller does not directly access the true system state. Instead, the controller receives measurements from sensors that may contain noise, degradation, or other measurement uncertainty. The sensor measurement can be represented as

where $x(t)$ denotes the true system state, $h(\cdot)$ represents the measurement function, $y(t)$ is the measured output, and $v(t)$ represents measurement noise.

For the robotic manipulator considered in this study, the state vector can be defined as

$$x(t) = [q(t)^T, \dot{q}(t)^T]^T$$

where $q(t)$ and $\dot{q}(t)$ are the joint positions and velocities, respectively.

For a direct joint-position measurement model, the measurement equation can be simplified to

$$y(t) = q(t) + v(t)$$

where $v(t)$ represents sensor noise.

To represent degraded sensing conditions, an additional measurement uncertainty term can be introduced:

$$y(t) = q(t) + v(t) + f_s(t)$$

where $f_s(t)$ represents a sensor degradation or fault component. depending on the experimental scenario, this term may represent biased measurements, increased measurement noise, intermittent degradation, or other controlled sensor imperfections.

The sensor model therefore provides a mechanism for evaluating the controller under progressively more challenging measurement conditions. The experiments will consider nominal measurements, noisy measurements, and degraded or uncertain measurements separately before evaluating their interaction with communication-induced uncertainty.

The specific statistical properties and degradation profiles of $v(t)$ and $f_s(t)$ will be defined during the experimental-design stage so that the selected models are consistent with the final simulation and validation framework.

3.3 Communication Network Model

The control loop is assumed to operate through a communication network connecting the sensing and control components. Consequently, the information available to the controller may be affected by communication delay and packet loss.

The received measurement can be represented as

$$y_r(t) = y(t - \tau(t))$$

where $y(t)$ is the transmitted measurement and $\tau(t)$ represents the time-varying communication delay.

To model packet loss, a binary variable can be introduced:

where $\gamma(t) = 1$ indicates successful reception of a transmitted measurement and $\gamma(t) = 0$ indicates a packet loss. The received signal can therefore be represented as

$$y_r(t) = \gamma(t)y(t - \tau(t))$$

When a packet is lost, the controller must operate using the available previous information or an appropriate state-estimation mechanism.

The communication model will allow different network conditions to be evaluated independently. These conditions will include nominal communication, fixed or time-varying delay, packet loss, and combinations of delay and packet loss. The corresponding parameters will be varied systematically during the simulation experiments.

Communication uncertainty will subsequently be considered together with sensor uncertainty and system disturbances. This enables the proposed framework to be evaluated under both isolated and simultaneous uncertainty conditions rather than treating the communication network as an independent subsystem.

3.4 Uncertainty/Disturbance Model

The robotic control system is subject to multiple uncertainty sources that may affect the accuracy and robustness of the closed-loop response. These sources are divided into three main categories: system uncertainty, sensor uncertainty, and communication uncertainty.

System uncertainty is represented by the difference between the nominal robot model and the actual system dynamics. It may include uncertain physical parameters, unmodelled dynamics, friction effects, and external disturbances. The disturbance term introduced in the robot dynamics is represented as

$$d(t) = d_m(t) + d_e(t)$$

where $d_m(t)$ represents model-related uncertainty and $d_e(t)$ represents external disturbances. Sensor uncertainty is represented by the measurement noise and degradation terms introduced in Section 3.2:

$y(t) = q(t) + v(t) + f_s(t)$ where $v(t)$ represents measurement noise and $f_s(t)$ represents sensor degradation or fault effects.

Communication uncertainty is represented through the time-varying delay and packet-loss process defined in Section 3.3. The resulting information available to the controller therefore depends on both the quality of the sensor measurement and the condition of the communication channel.

For the proposed research, these uncertainty sources are evaluated both independently and simultaneously. This distinction is important because a controller that performs adequately

under a single uncertainty source may experience substantially different behavior when multiple uncertainty sources interact within the same feedback loop.

The uncertainty framework will therefore be used to define the experimental scenarios and to provide the input conditions for the subsequent learning-based uncertainty estimation and robust adaptive control stages.

3.5 Control Objective

The primary objective is to achieve accurate trajectory tracking while maintaining acceptable control performance in the presence of sensor uncertainty, communication uncertainty, system uncertainty, and external disturbances.

Given the desired trajectory $q^d(t)$, the controller is required to minimize the tracking error

$$e(t) = q(t) - q_d(t)$$

While ensuring that the applied control input remains within the practical operating limits of the robotic system.

The control objective can therefore be expressed as

$$e(t) \rightarrow 0$$

Subject to the presence of uncertain dynamics, measurement imperfections, communication delay, packet loss, and external disturbances.

subject to the presence of uncertain dynamics, measurement imperfections, communication delay, packet loss, and external disturbances.

In addition to tracking accuracy, the proposed framework will be evaluated according to several performance objectives:

- tracking error;
- maximum tracking deviation;
- settling time;
- overshoot;
- control effort;
- disturbance rejection;
- robustness under increasing uncertainty;
- computational cost.

The principal research objective is not to assume that learning-based uncertainty estimation provides an improvement, but to experimentally determine whether incorporating such information into the robust adaptive controller produces measurable benefits compared with

appropriate baseline controllers under equivalent operating conditions. These evaluation criteria are consistent with the experimental objectives defined for the study.

3.6 Formal Problem Statement

Consider a nonlinear robotic manipulator described by the dynamics in (1), where the controller receives imperfect and potentially delayed measurements according to the sensor and communication models defined in Sections 3.2 and 3.3.

The control problem is to design a robust adaptive controller augmented with a learning-based uncertainty estimation mechanism such that the manipulator tracks a desired trajectory $qd(t)$ despite the simultaneous presence of:

- uncertain and unmodelled robot dynamics;
- external disturbances;
- sensor noise and degradation;
- communication delay;
- packet loss.

The proposed framework can be represented conceptually as

where $N(t)$ denotes relevant network conditions and $u(t)$ represents the uncertainty information estimated by the learning component.

The research problem is therefore to determine whether incorporating learned uncertainty information into the robust adaptive control mechanism can improve closed-loop performance under combined uncertainty conditions relative to appropriate baseline controllers.

The resulting problem will be evaluated experimentally through MATLAB/Simulink-based simulations and controlled uncertainty scenarios, with the final experimental configuration determined in Section 5.

Proposed Methodology

4.1 Overall Architecture

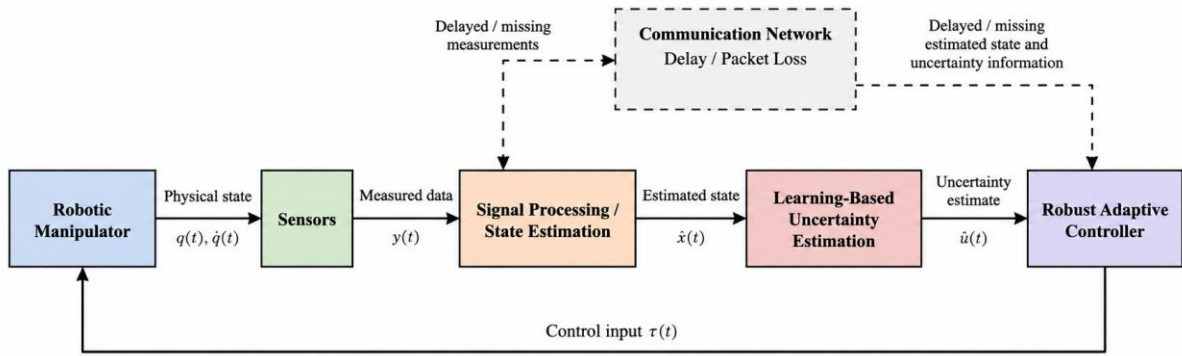
The proposed control framework consists of five main stages: sensing, signal processing and state estimation, learning-based uncertainty estimation, robust adaptive control, and robotic actuation. The sensor subsystem provides measurements of the robotic manipulator. These measurements are processed to reduce the effects of noise and to obtain the system state required for feedback control. The estimated state and relevant measurement information are then provided to the learning-based uncertainty estimation stage.

The learning component is intended to estimate information related to the uncertainty affecting the closed-loop system. Its inputs may include estimated system states, measurement characteristics, tracking information, and communication-related variables such as delay and packet-loss indicators. The output is an estimate or representation of the current uncertainty level.

The robust adaptive controller remains the primary control mechanism. Rather than replacing the controller with a learned control policy, the learning component provides uncertainty information that can be incorporated into the controller's adaptation or robustness mechanism. The resulting control input is then applied to the robotic manipulator.

Communication uncertainty is introduced between the relevant sensing, estimation, and control stages. This allows the architecture to represent delayed or missing information within the feedback loop.

Figure 1. Overall architecture of the proposed learning-enhanced robust adaptive control framework.



4.2 Signal Processing and State Estimation

The sensor measurements obtained from the robotic manipulator may contain noise and other measurement imperfections that can adversely affect the control process. Therefore, a signal-processing and state-estimation stage is included before the learning-based uncertainty estimation and control stages.

The measured signal is first processed to reduce high-frequency measurement noise and other undesired components. The processed measurements are then used to estimate the system state required by the controller. The estimated state can be represented as

$$x_{hat(t)} = [q_{hat(t)}^T, \{\dot{q}\}_{hat(t)}^T]^T$$

where $q_{hat}(t)$ and $\dot{q}_{hat}(t)$ denote the estimated joint positions and velocities, respectively.

The state-estimation stage provides an estimate of the variables that are required for trajectory tracking and uncertainty estimation. In particular, the estimated state is used to construct the tracking error and to provide relevant input information to the learning-based uncertainty estimator.

Because the measured information may be affected by both sensor noise and communication imperfections, the estimator must operate on the information that is actually available within the networked feedback loop. Delayed or missing measurements may therefore affect the quality of the estimated state.

At the current stage, the specific state-estimation algorithm is not fixed. The final estimator will be selected based on the computational requirements of the robotic system, the characteristics of the simulated sensor noise, and the communication uncertainty model. The selected method will subsequently be implemented and evaluated under both nominal and degraded measurement conditions.

The performance of the state-estimation stage will be considered as part of the overall control architecture rather than as an independent optimization objective. Its primary role is to provide sufficiently reliable state information for the subsequent uncertainty-estimation and robust adaptive control stages.

4.3 Learning-Based Uncertainty Estimation

The learning-based component is introduced to estimate uncertainty information from data collected during operation of the robotic system. Unlike a controller that directly learns the control policy, the proposed approach assigns the learning component a specific role within the overall control architecture: estimating uncertainty that can subsequently be used by the robust adaptive controller.

The estimator receives information describing the current operating condition of the robot. A general input vector can be represented as

$$z(t) = [x_{hat}(t), e(t), \dot{e}(t), r(t)]^T$$

where $x_{hat}(t)$ represents the estimated system state, $e(t)$ and $\{\dot{e}\}(t)$ represent the tracking errors, and $r(t)$ represents additional information related to the measurement or communication conditions.

The learning model can then be expressed in a general form as

$u_{hat}(t) = f_{\theta}(\cdot)$ where $f_{\theta}(\cdot)$ represents the learning model with parameters θ , and $u_{hat}(t)$ represents the estimated uncertainty information.

The uncertainty estimate may contain information related to the magnitude or characteristics of model uncertainty, disturbances, sensor degradation, or communication-induced effects. The exact output representation will depend on the final learning architecture and the uncertainty variables selected for the experimental study.

The purpose of the estimator is not to replace the analytical control mechanism. Instead, its output is provided to the robust adaptive controller so that the controller can account for changes in the uncertainty conditions. This separation between uncertainty estimation and control is consistent with learning-based robust-control approaches in which learned information is incorporated into an existing stabilizing control structure [5]. Neural and learning-based adaptive methods have similarly been investigated for representing unknown dynamics and disturbances in robotic systems [1], [17].

A key part of the methodology will be determining how the learning model is trained and what constitutes the uncertainty target. Because the uncertainty sources in this study can be controlled within simulation, the experimental environment can provide known disturbance, sensor, and communication conditions from which suitable training and evaluation data can be generated.

The learning component will therefore be evaluated not only according to the accuracy of its uncertainty estimates, but also according to whether those estimates provide a measurable benefit when incorporated into the robust adaptive controller. This distinction is important because an accurate uncertainty prediction is not necessarily sufficient to improve closed-loop control performance.

4.4 Robust Adaptive Controller

The robust adaptive controller represents the primary control mechanism of the proposed framework. Its purpose is to maintain satisfactory trajectory tracking despite uncertainty in the robotic dynamics, external disturbances, imperfect state information, and communication-related effects.

Unlike the learning component, the controller is not intended to learn the complete control policy from data. Instead, it remains a model-based robust adaptive control mechanism whose parameters or robustness characteristics can be adjusted according to the available uncertainty information. This separation preserves a genuine control-theoretic component within the proposed architecture.

The controller receives the estimated system state, the desired trajectory, the tracking error, and the uncertainty information generated by the learning-based estimator. In general form, the control input can be expressed as

$$\tau(t) = pi\left(x_{hat(t)}, q_{d(t)}, dot(q_{d(t)}), u_{hat(t)}, \theta_{(a(t))}\right)$$

where $pi(.)$ denotes the robust adaptive control law, $u_{hat(t)}$ represents the estimated uncertainty, and $\theta_{a(t)}$ represents the adaptive controller parameters.

The adaptive mechanism is intended to compensate for changes in the effective system dynamics or uncertainty conditions during operation. The robust component provides additional protection against uncertainty and disturbances that cannot be accurately represented by the adaptive model. This combination is motivated by existing research on adaptive, disturbance-observer-based, and robust control methods for uncertain robotic manipulators [7]–[12].

A central design requirement is that the learning-based uncertainty estimator should complement rather than replace the robust adaptive controller. The controller must therefore remain capable of operating when the learning component is removed or when its uncertainty estimate is inaccurate. This property will allow the learning contribution to be examined through the planned ablation experiments.

The exact robust adaptive control law will be selected after the final system model and experimental requirements have been established. Candidate approaches include adaptive sliding-mode, robust adaptive nonlinear, and observer-assisted adaptive control structures. The final selection will be based on theoretical suitability, implementation complexity, computational cost, and compatibility with the uncertainty-estimation mechanism.

The controller will subsequently be evaluated against the selected baseline controllers under identical uncertainty conditions. This comparison will determine whether the incorporation of learned uncertainty information provides a measurable improvement in tracking accuracy, robustness, disturbance rejection, and control effort rather than assuming such an improvement in advance.

4.5 Integration of Learning and Control

The learning-based uncertainty estimator and the robust adaptive controller are integrated within the same closed-loop control architecture. The estimated system state, tracking error, and relevant sensing and communication information are first processed by the learning component to obtain an estimate of the current uncertainty.

The estimated uncertainty is then provided to the robust adaptive controller and used to modify its adaptation or robustness mechanism. The resulting control input is calculated from both the conventional control information and the learned uncertainty information.

The overall control relationship can be represented as

$$\tau(t) = pi\left(x_{hat(t)}, q_{d(t)}, dot(q_{d(t)}, u_{hat(t)}, \theta_{a(t)})\right)$$

where $\hat{q}$ is the estimated state, $\hat{\Sigma}$ is the learned uncertainty estimate, and θ represents the adaptive controller parameters.

This architecture preserves a clear separation between estimation and control. The learning component estimates uncertainty, while the robust adaptive controller remains responsible for generating the control action. This also makes it possible to evaluate the contribution of the learning component independently through ablation experiments.

The integrated system will be evaluated under nominal conditions and under progressively increasing levels of sensor, communication, dynamic, and disturbance uncertainty. Performance will then be compared with baseline controllers operating under the same conditions.

4.6 Stability/Robustness

The proposed control framework is intended to maintain stable and reliable closed-loop operation in the presence of uncertainty. The robust adaptive controller provides the primary mechanism for handling model uncertainty and external disturbances, while the learning-based component provides additional uncertainty information.

The stability analysis will be formulated with respect to the tracking error defined as

$$e(t) = q(t) - q_d(t)$$

and its corresponding closed-loop dynamics. A Lyapunov-based analysis may be used to evaluate whether the selected controller maintains bounded system states and tracking errors under the considered uncertainty conditions.

The learning-based uncertainty estimator will not be assumed to be perfectly accurate. Its estimation error will therefore be considered as an additional uncertainty term in the control analysis. The robust component of the controller should provide sufficient tolerance to estimation errors so that inaccuracies in the learned uncertainty estimate do not directly lead to loss of closed-loop robustness.

The final stability conditions will depend on the specific adaptive and robust control law selected in the subsequent methodology-development stage. Consequently, the mathematical stability proof will be finalized after the exact controller structure and learning-based uncertainty representation have been defined.

The experimental evaluation will complement the theoretical analysis by examining tracking-error boundedness, transient response, disturbance rejection, and performance degradation as the magnitude of sensor and communication uncertainty is increased.

4.7 Computational Complexity

The computational requirements of the proposed framework arise primarily from three components: state estimation, learning-based uncertainty estimation, and robust adaptive control. Each component must operate within a time scale compatible with the closed-loop dynamics of the robotic manipulator.

The computational cost of the state-estimation stage depends on the selected filtering or observer method and the dimension of the system state. The learning-based uncertainty estimator additionally introduces the computational cost associated with evaluating the trained model. This cost depends on the selected model architecture, number of parameters, and input dimension.

The robust adaptive controller requires real-time evaluation of the control law and the associated adaptation mechanism. Its computational requirements depend on the mathematical complexity of the controller and the number of adaptive parameters.

The total computational requirement can therefore be considered as the combined cost of state estimation, uncertainty estimation, and control computation. The final implementation will report the execution time and other relevant computational measurements required to determine whether the proposed framework is practical for the selected simulation and control sampling rate.

Computational performance will also be included in the comparative evaluation. The proposed method will be compared with the selected baseline controllers under equivalent simulation conditions so that any performance improvement can be considered together with the additional computational cost introduced by the learning component.

Experimental Methodology

5.1 Simulation Environment

The proposed control framework will initially be evaluated in a numerical simulation environment using MATLAB and Simulink. The simulation environment will provide a controlled platform for implementing the robotic manipulator dynamics, sensor model, communication network model, learning-based uncertainty estimator, and robust adaptive controller.

The robotic manipulator model defined in Section 3 will be implemented together with the desired reference trajectories. Sensor noise and degradation will be introduced through the measurement model, while communication delay and packet loss will be incorporated into the feedback path according to the network model.

The simulation environment will allow the uncertainty sources to be controlled independently and in combination. This will make it possible to evaluate the response of the proposed controller under nominal conditions, individual uncertainty conditions, and simultaneous uncertainty conditions using the same robot model and parameter configuration.

The learning-based uncertainty estimator and the robust adaptive controller will be implemented within the same simulation framework. The baseline controllers will be evaluated under identical conditions to ensure that the comparison is based on equivalent reference trajectories, disturbances, communication conditions, and simulation parameters.

The final simulation parameters, sampling time, robot configuration, learning-model architecture, and uncertainty ranges will be specified once the exact controller and experimental design have been finalized.

5.2 Robotic Platform and Parameters

A 2-DOF robotic manipulator will be used as the primary simulation platform for evaluating the proposed control framework. The platform provides a nonlinear dynamic system with coupled joint motion while remaining sufficiently manageable for repeated numerical experiments and controlled uncertainty studies.

The manipulator model will be defined by the physical and dynamic parameters required by the robot equation in Section 3.1. These parameters will include link lengths, link masses, moments of inertia, gravitational acceleration, friction characteristics, and actuator constraints. The nominal parameter set will be used to construct the reference model available to the controller.

To evaluate robustness against model uncertainty, selected physical parameters of the simulated manipulator will be varied from their nominal values during the experiments. These variations will represent changes in the actual robot dynamics that are not fully captured by the nominal controller model.

The desired joint trajectories and initial conditions will be kept consistent across the baseline and proposed controllers. The same robot model, reference trajectories, actuator constraints, disturbance profiles, and communication conditions will be used for comparative experiments.

The final numerical values of the robotic parameters will be specified after the 2-DOF manipulator model has been selected and implemented in MATLAB/Simulink. The complete parameter set will be documented to ensure that the simulation experiments are reproducible.

5.3 Baseline Controllers

To evaluate the contribution of the learning-based uncertainty estimation mechanism, the proposed framework will be compared with progressively more advanced control

configurations. All controllers will operate under the same robotic model, reference trajectories, disturbances, sensor conditions, and communication conditions.

The baseline configurations will include:

Baseline 1 — Conventional Controller:

A conventional feedback controller will provide the basic performance reference under nominal and uncertain operating conditions.

Baseline 2 — Robust Controller:

A robust controller will be evaluated to determine the performance obtained through robustness mechanisms without adaptive or learning-based uncertainty estimation.

Baseline 3 — Robust Adaptive Controller:

An adaptive mechanism will be added to account for changes in system dynamics and uncertainty. This controller will serve as the primary non-learning baseline.

Proposed Controller — Learning-Enhanced Robust Adaptive Controller:

The selected robust adaptive controller will be augmented with the learning-based uncertainty estimator described in Section 4.3.

The final mathematical form of the robust adaptive controller and the conventional baseline controller will be selected after the controller-design stage. The selection will be based on relevance to the identified research gap, theoretical properties, implementation feasibility, and suitability for MATLAB/Simulink experimentation.

The comparison is designed to isolate the contribution of the learning component. In particular, the comparison between the robust adaptive baseline and the proposed controller will determine whether the learned uncertainty information provides an additional measurable benefit.

5.4 Experimental Scenarios

The experimental study will evaluate the control framework under progressively more challenging operating conditions. The same reference trajectory, robot parameters, controller settings, and simulation duration will be used across the compared controllers to ensure a consistent basis for evaluation.

The first scenario will represent **nominal operation**, in which the robot model, sensor measurements, and communication channel operate without intentionally introduced uncertainties. This scenario establishes the baseline tracking performance of each controller.

The second scenario will introduce **sensor noise** into the measurements. The noise level will be controlled to evaluate how measurement uncertainty affects state estimation and trajectory tracking.

The third scenario will consider **sensor degradation or fault conditions**. Controlled measurement disturbances will be introduced to represent degraded sensing conditions and evaluate the ability of the controllers to maintain tracking performance.

The fourth scenario will introduce **communication delay** into the feedback loop. Different delay conditions may be evaluated to determine the sensitivity of each controller to outdated measurement information.

The fifth scenario will introduce **packet loss**. Measurements will be intermittently unavailable to the controller according to the selected packet-loss model. The resulting degradation in tracking performance will be evaluated.

The sixth scenario will introduce **external disturbances** to the robotic system. These disturbances will be applied to evaluate disturbance-rejection capability and robustness.

The seventh scenario will introduce **changing or uncertain robot dynamics**, such as variations in selected physical parameters relative to the nominal model. This scenario will evaluate the adaptive capability of the controllers.

The final scenario will combine multiple uncertainty sources, including sensor uncertainty, communication effects, dynamic uncertainty, and external disturbances. This combined scenario is particularly important because it represents the main operating condition investigated by this research.

An ablation experiment will also be performed by removing the learning-based uncertainty estimation component while keeping the remaining controller structure unchanged. The comparison will be used to isolate the contribution of the learning component.

The final numerical values for noise intensity, sensor degradation, communication delay, packet-loss probability, disturbance magnitude, and parameter variation will be selected during implementation and reported explicitly to ensure reproducibility.

5.5 Evaluation Metrics

The performance of the investigated controllers will be evaluated using quantitative metrics that capture tracking accuracy, transient response, robustness, control effort, and computational requirements.

The primary tracking-performance metric will be the root mean square error (RMSE), calculated from the trajectory-tracking error over the simulation interval. Maximum tracking error will also be considered to identify the largest deviation from the desired trajectory.

Transient performance will be evaluated using settling time and overshoot. These metrics will indicate how quickly the controller responds to changes in the reference trajectory and how effectively it limits transient deviations.

Control effort will be evaluated from the magnitude of the applied control input. This metric is important because improved tracking performance obtained through excessively large control actions may not represent a practical improvement.

Disturbance-rejection performance will be evaluated by measuring the degradation in tracking accuracy after external disturbances are introduced. Robustness will additionally be assessed by comparing controller performance as the magnitude of sensor, communication, and dynamic uncertainty is progressively increased.

The computational cost of each controller will also be measured. The proposed learning-enhanced framework will therefore be evaluated not only according to its tracking performance but also according to the additional computational requirements introduced by the learning component.

The principal metrics considered in the experimental study are:

- tracking RMSE;
- maximum tracking error;
- settling time;
- overshoot;
- control effort;
- disturbance-rejection performance;
- performance degradation under increasing uncertainty;
- computational cost.

These metrics correspond to the evaluation criteria identified in the initial research plan.

5.6 Implementation Details

The proposed framework will be implemented in MATLAB/Simulink to provide a reproducible simulation environment for the robotic manipulator, sensor model, communication network, learning-based uncertainty estimator, and control algorithms.

The implementation will use the same simulation settings for all controller configurations. These settings will include the robot model and parameters, reference trajectories, sampling configuration, sensor-noise model, sensor-degradation model, communication-delay model, packet-loss model, disturbance signals, and dynamic-parameter variations.

The learning-based uncertainty estimator will be trained using data generated from the simulated robotic system under the uncertainty conditions defined in Section 5.4. The training and evaluation datasets will be separated so that the performance of the estimator and the resulting controller can be evaluated on conditions not directly used during training.

The implementation will use fixed experimental configurations for each comparison. Where stochastic noise, packet loss, or randomly generated disturbances are used, multiple runs will

be performed using controlled random seeds. Mean performance and variability across repeated trials will be reported where appropriate.

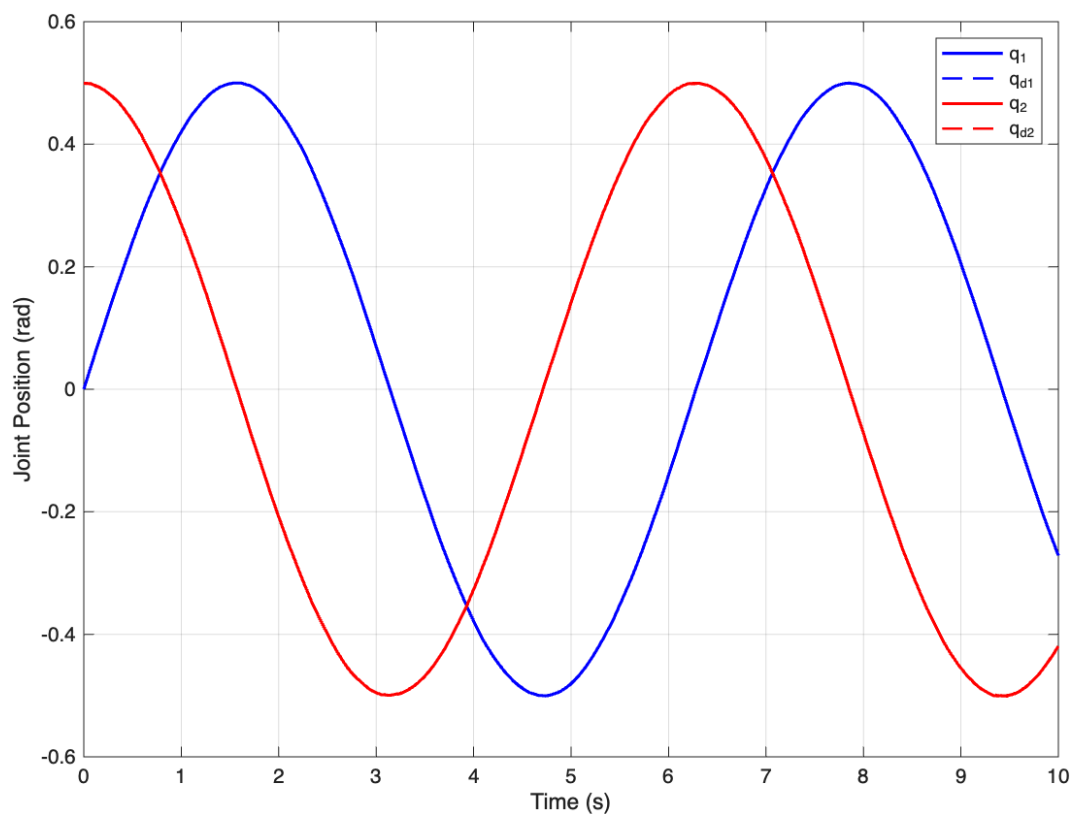
All controller parameters, learning-model parameters, simulation settings, and uncertainty configurations will be documented. The MATLAB/Simulink implementation will be organized so that the experiments can be reproduced by executing predefined simulation and analysis scripts.

The final implementation details, including the MATLAB version, solver configuration, sampling time, learning architecture, training procedure, dataset size, and numerical uncertainty ranges, will be specified after the controller and learning method have been finalized.

Results / Findings

6.1 Nominal Conditions

Figure 2. Joint trajectory tracking under nominal operating conditions.



Under nominal operating conditions, the adaptive sliding-mode controller was first evaluated without communication delay, packet loss, sensor degradation, or external disturbances. The desired joint trajectories were sinusoidal reference signals for the two-degree-of-freedom robotic manipulator.

As shown in Figure 2, the actual joint trajectories closely follow the corresponding reference trajectories, demonstrating accurate tracking under nominal conditions. This experiment provides the baseline control performance against which the subsequent uncertainty scenarios are compared.

6.2 Sensor Noise

Figure 3. Joint trajectory tracking under sensor noise.

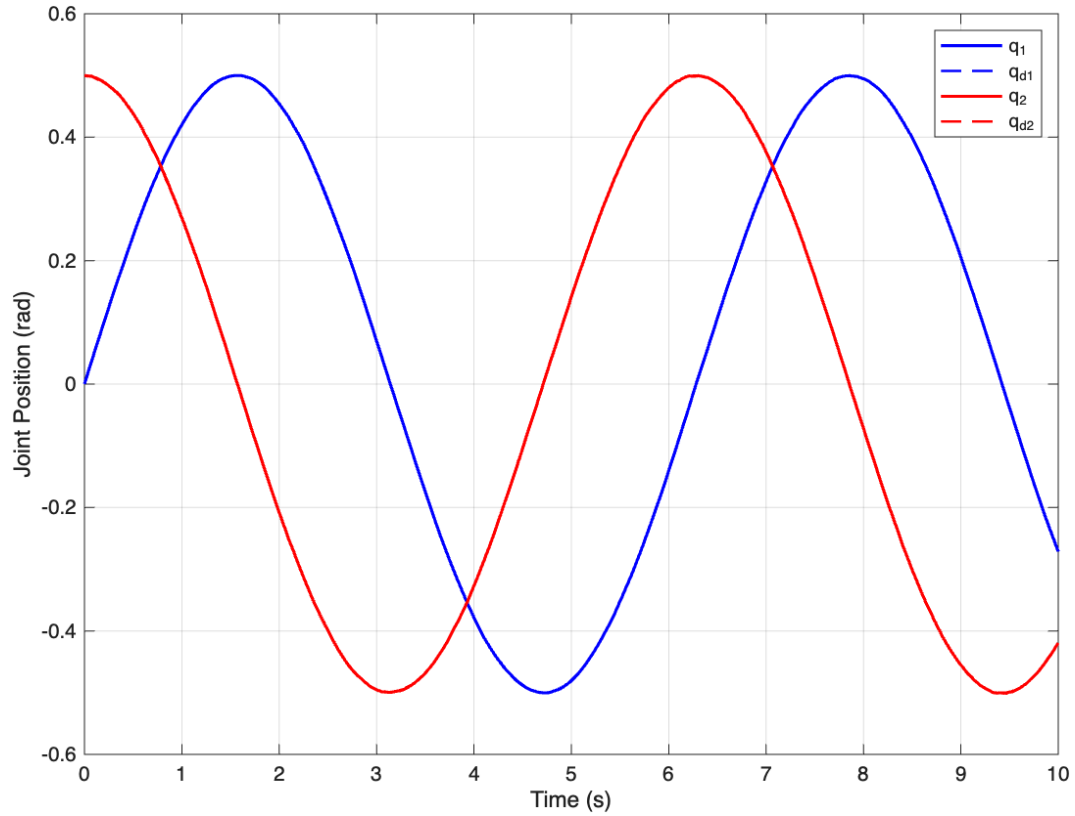
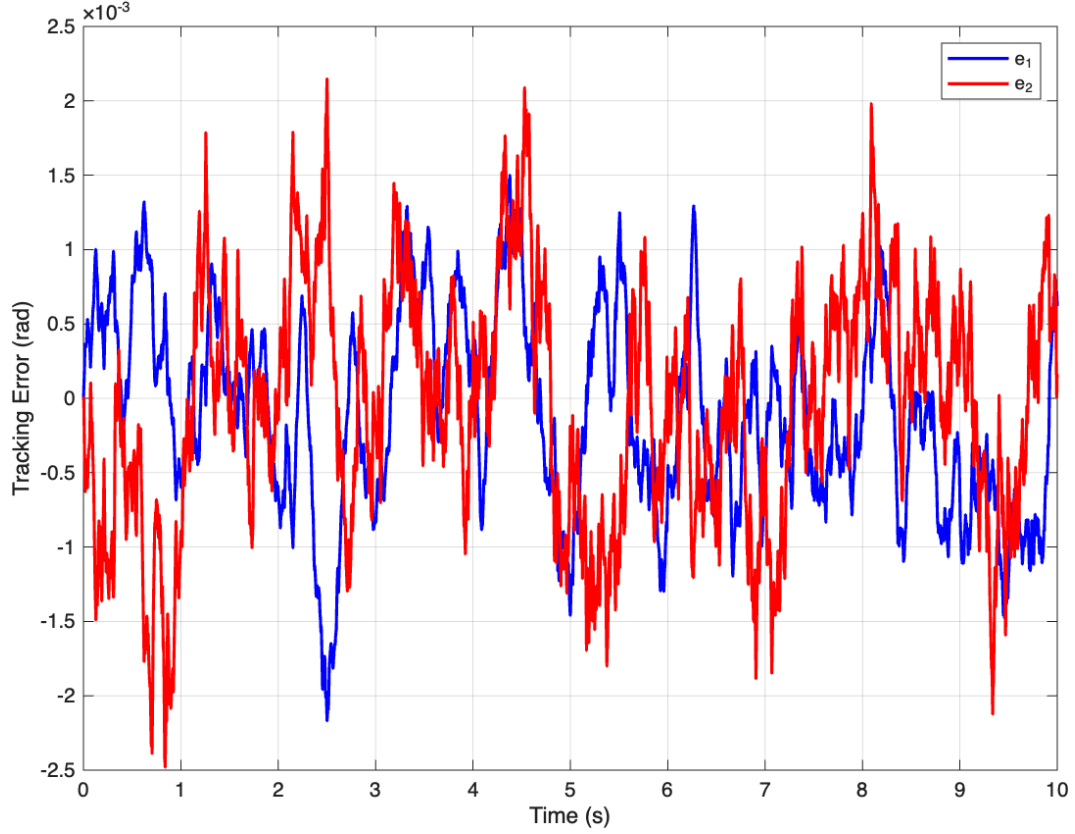


Figure 4. Tracking error under sensor noise.



Sensor noise was introduced into the joint-position and joint-velocity measurements to evaluate the tracking performance of the adaptive sliding-mode controller under imperfect sensing conditions. The noise standard deviations were set to 0.01 rad for joint position and 0.02 rad/s for joint velocity.

As shown in Figure 3, the controller maintains trajectory tracking despite the presence of measurement noise. Figure 4 shows the corresponding tracking errors for the two joints.

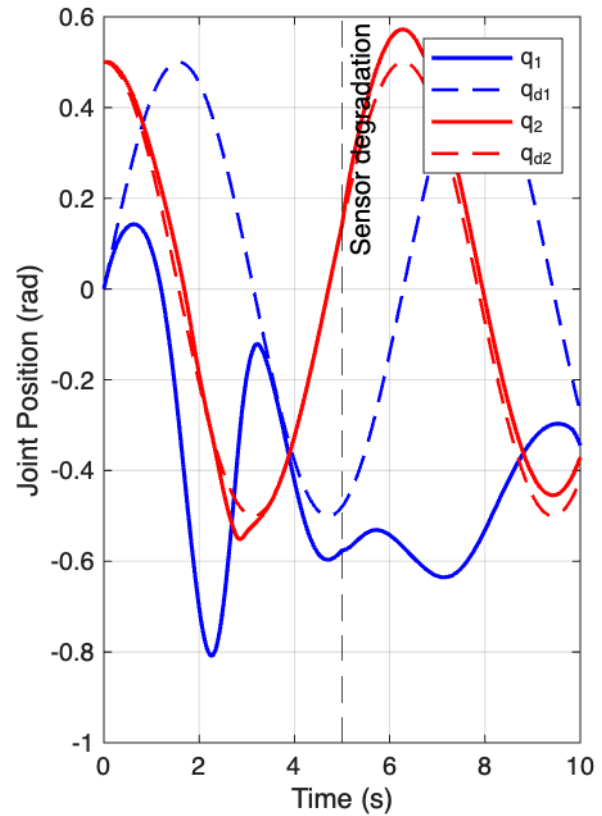
The obtained RMSE values were **0.000658 rad for q_1** and **0.000768 rad for q_2** . The maximum absolute tracking errors were **0.002168 rad for q_1** and **0.002481 rad for q_2** .

6.3 Sensor Uncertainty / Degradation

Sensor degradation was introduced by applying a constant measurement bias after 5 s of operation. The bias values were set to +0.05 rad for the first joint and -0.05 rad for the second joint.

The resulting tracking performance is shown in Figure 5. The degradation produces a noticeable change in the tracking behaviour, particularly for the first joint. The measured RMSE values were 0.630352 rad for q_1 and 0.043255 rad for q_2 . The corresponding maximum absolute tracking errors were 1.206938 rad and 0.075929 rad, respectively.

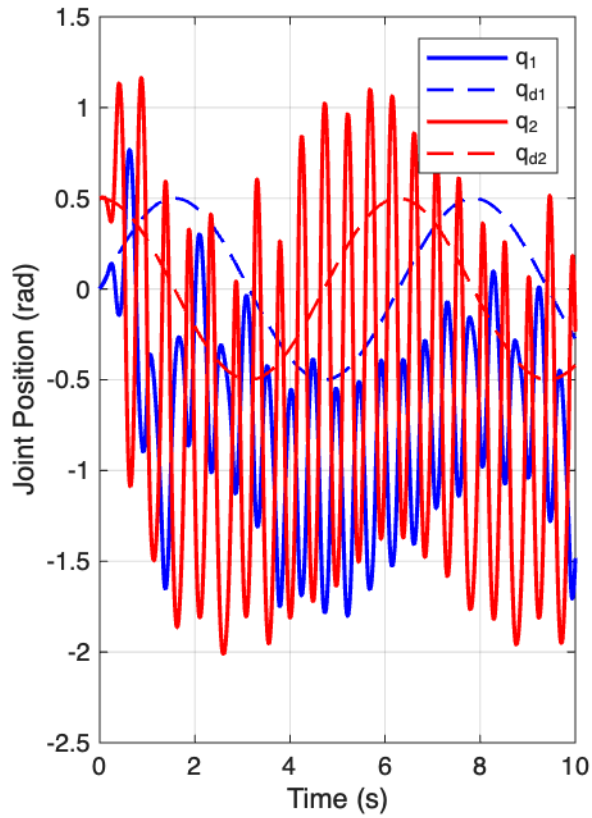
Figure 5. Joint trajectory tracking under sensor noise.



6.4 Communication Delay

A communication delay of 100 ms was introduced into the feedback loop to evaluate the effect of delayed state information on the tracking performance. The delayed measurements caused a significant degradation in trajectory tracking compared with the nominal and sensor-noise scenarios. Under the 100 ms communication delay condition, the RMSE values were **0.927737 rad for q_1** and **1.034056 rad for q_2** . The maximum absolute tracking errors were **2.146842 rad for q_1** and **1.904390 rad for q_2** .

Figure 6. Joint trajectory tracking under 100 ms communication delay.

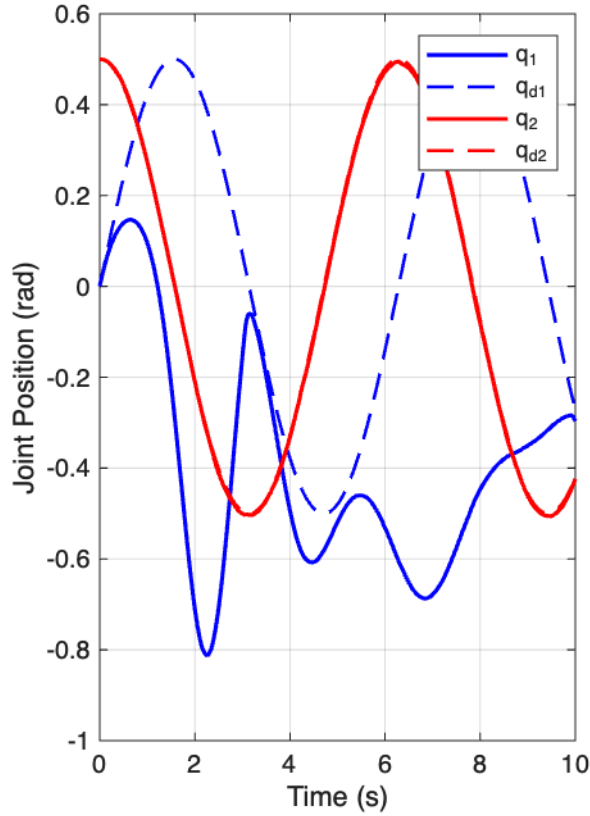


6.5 Packet Loss

A packet-loss probability of 10% was introduced into the feedback communication channel to evaluate the effect of intermittently unavailable measurements on trajectory tracking. When a packet was lost, the controller retained the most recently received measurement.

Under this condition, the RMSE values were 0.624917 rad for $q1$ and 0.006313 rad for $q2$. The maximum absolute tracking errors were 1.213158 rad for $q1$ and 0.012446 rad for $q2$.

Figure 7. Joint trajectory tracking under 10% packet loss.



6.6 External Disturbances

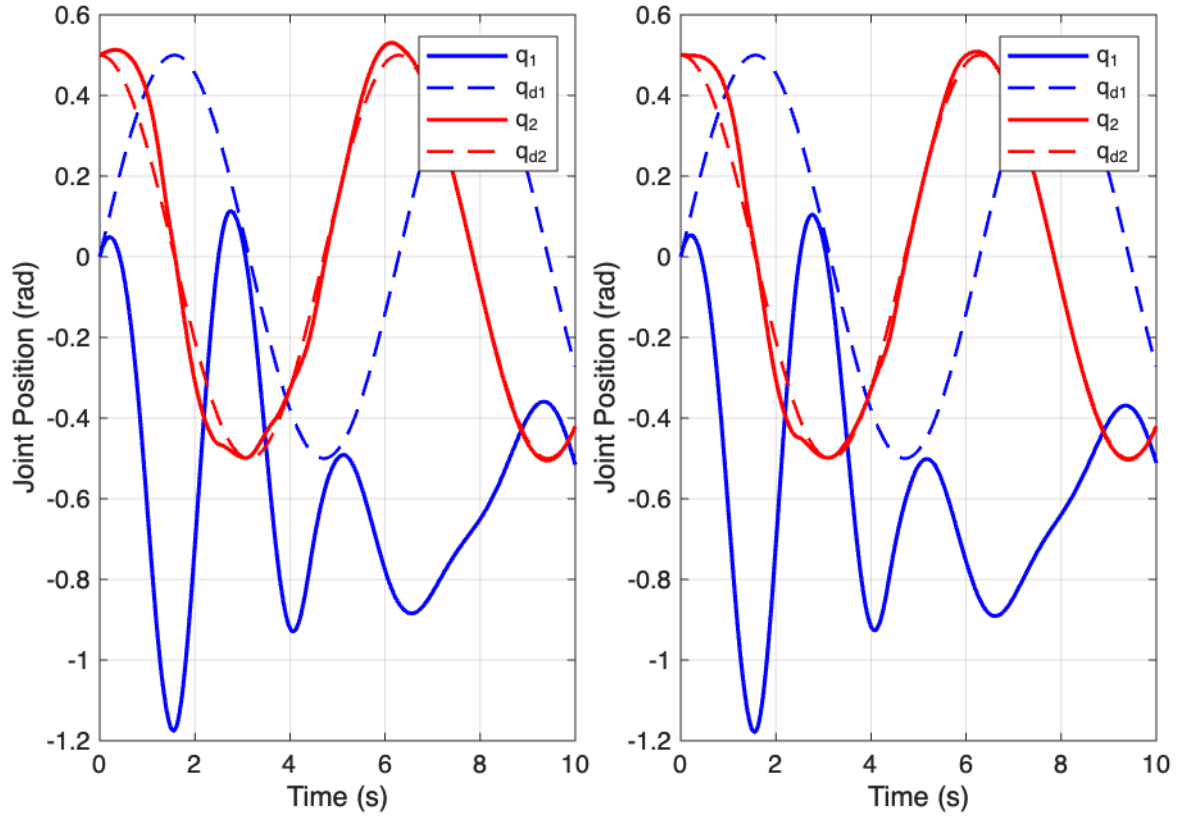
To evaluate the behavior of the controllers under physical uncertainty, the nominal robot parameters were deliberately modified and external disturbances were introduced. The actual manipulator parameters were set to $m1=1.90$ kg and $m2=2.20$ kg, while the nominal controller model used $m1=1.70$ kg and $m2=2.00$ kg. In addition, time-varying external disturbances were applied to both joints.

Under these conditions, the ASMC achieved RMSE values of **0.798392 rad for $q1$** and **0.039690 rad for $q2$** . The maximum absolute tracking errors were **1.678349 rad** and **0.124463 rad**, respectively.

The corresponding ASMC + RBFNN experiment produced RMSE values of **0.798705 rad for $q1$** and **0.046209 rad for $q2$** , with maximum absolute errors of **1.675247 rad** and **0.133696 rad**. In this experiment, the learning-based uncertainty estimator did not produce an improvement in tracking RMSE. The results therefore indicate that the contribution of the learning component is not uniform across all uncertainty conditions.

Figure 8. Joint trajectory tracking using ASMC under external disturbances and changing dynamics.

Figure 9. Joint trajectory tracking using ASMC + RBFNN under external disturbances and changing dynamics.



6.7 Combined Uncertainty

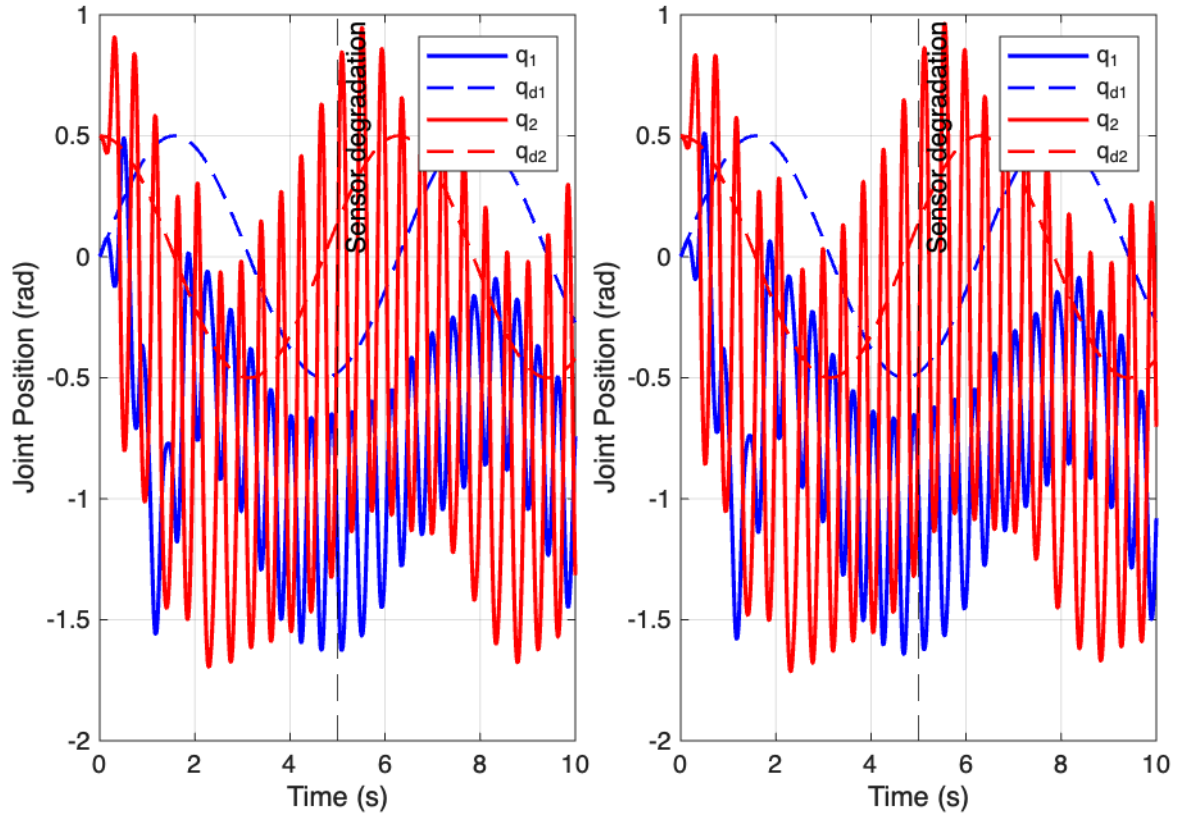
The principal experiment combined sensor noise, communication delay, packet loss, uncertain robot dynamics, and external disturbances within the same closed-loop system. The sensor noise standard deviations were set to 0.01 rad for joint position and 0.02 rad/s for joint velocity, the communication delay was set to 100 ms, and the packet-loss probability was 10%.

To reduce dependence on a single stochastic realization, the experiment was repeated using 10 different random seeds. The mean and standard deviation of the tracking RMSE are presented below.

Metric	ASMC	ASMC + RBFNN
RMSE $q1$ [rad]	0.933173 ± 0.005594	0.883266 ± 0.010480
RMSE $q2$ [rad]	1.041890 ± 0.005282	0.852563 ± 0.002524

The corresponding mean RMSE reduction was **5.35% for $q1$** and **18.17% for $q2$** . These results provide evidence that the learning-based uncertainty estimator can reduce tracking error under the combined uncertainty configuration investigated in this study.

Figure 9. Comparison of ASMC and ASMC + RBFNN under combined uncertainty.



6.8 Ablation Study

An ablation comparison was performed by removing the learning-based uncertainty estimation component while keeping the remaining robust adaptive control structure unchanged. The ASMC therefore represents the non-learning baseline, while ASMC + RBFNN represents the proposed learning-enhanced configuration.

The combined-uncertainty experiment demonstrated a measurable reduction in RMSE when the RBFNN component was included. However, the additional sensor-degradation experiment produced different behavior. Across 10 seeds, the results were:

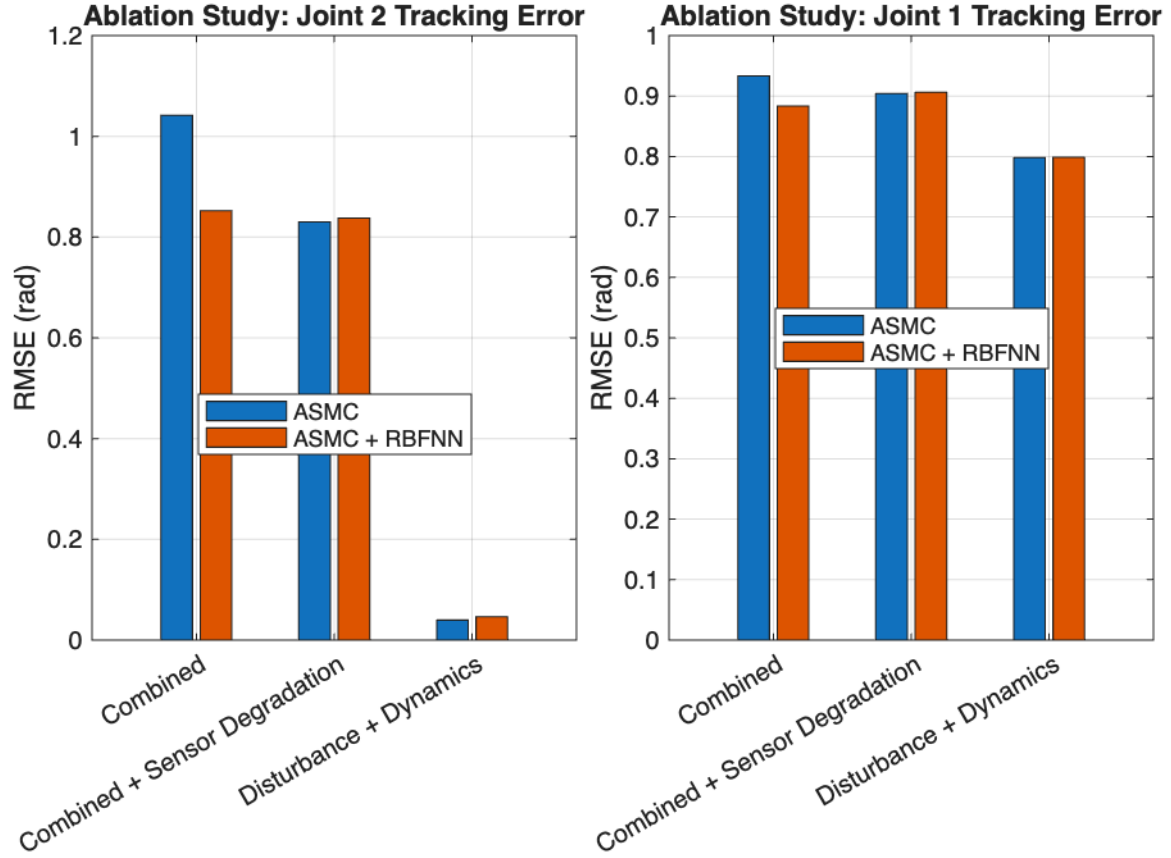
Metric	ASMC	ASMC + RBFNN
RMSE q_1 [rad]	0.904095 ± 0.008575	0.906200 ± 0.009096
RMSE q_2 [rad]	0.829856 ± 0.001640	0.837462 ± 0.003339

The corresponding changes in RMSE were **-0.23% for q_1** and **-0.92% for q_2** . Thus, the learning component did not provide an additional tracking benefit under the combined uncertainty and sensor-degradation configuration used in the experiment.

The ablation results indicate that the effect of the learning-based uncertainty estimator depends on the uncertainty conditions under which it is applied. In particular, the estimator

provided a measurable benefit in the combined networked-uncertainty experiment but not when sensor degradation was additionally introduced.

Figure 11. Ablation study of joint 1 and 2 tracking performances under different uncertainty conditions.



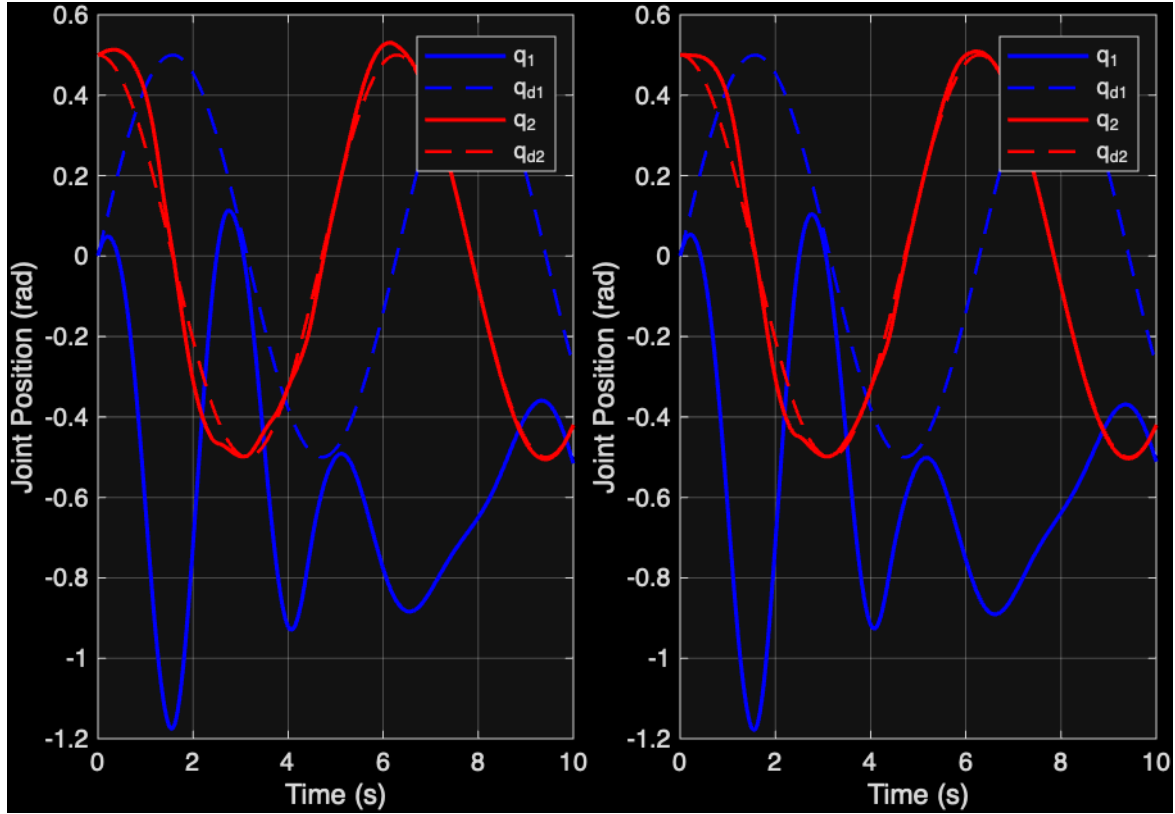
6.9 Computational Performance

The computational and control-related cost of the proposed framework was assessed through the control effort required during the combined uncertainty and sensor-degradation experiment. The control effort was calculated from the magnitude of the two control inputs over the complete simulation interval.

Across 10 random seeds, the mean control effort was $271.124316 \pm 0.257903 \text{ N}\cdot\text{m}\cdot\text{s}$ for ASMC and $271.173418 \pm 0.172300 \text{ N}\cdot\text{m}\cdot\text{s}$ for ASMC + RBFNN. The resulting change in control effort was $+0.02\%$.

These results indicate that the addition of the RBFNN produced only a small change in the total control effort for this experimental configuration. However, execution-time profiling and processor-level computational measurements were not recorded in the current experiments. Therefore, a quantitative conclusion regarding the computational runtime overhead of the learning component cannot yet be established.

Table 2. Control effort comparison between ASMC and ASMC + RBFNN.



Discussion

7.1 Interpretation of Results

The experimental results show that the effect of learning-based uncertainty estimation depends strongly on the type and combination of uncertainty present in the control loop.

Under the combined networked-uncertainty scenario, consisting of sensor noise, 100 ms communication delay, 10% packet loss, uncertain robot dynamics, and external disturbances, the ASMC + RBFNN configuration produced lower mean tracking RMSE than the ASMC baseline for both joints. Across 10 random seeds, the RMSE decreased by 5.35% for $q1$ and 18.17% for $q2$. This indicates that the learned uncertainty estimate can provide useful information to the robust adaptive controller under the particular networked conditions investigated.

However, the improvement was not observed when sensor degradation was additionally introduced. In that experiment, the ASMC + RBFNN configuration produced slightly higher RMSE values than the ASMC baseline, with changes of -0.23% for $q1$ and -0.92% for $q2$. The control effort changed by only 0.02%, indicating that the additional learning component did not materially alter the total control effort in this configuration.

The experiment involving changing dynamics and external disturbances produced a similar result. The RMSE obtained with the RBFNN was comparable to, but not lower than, the

ASMC baseline. This suggests that accurate uncertainty estimation alone does not necessarily translate into improved closed-loop tracking performance.

Overall, the results do not support the assumption that learning-based uncertainty estimation universally improves robust adaptive control. Instead, the results suggest that its usefulness is dependent on how closely the learned uncertainty representation corresponds to the uncertainty affecting the control input and on how that estimate is integrated into the control law.

7.2 Comparison with Existing Approaches

The experimental findings should be considered in the context of the control and learning approaches reviewed in Section 2. Existing work demonstrates that robust and adaptive control can provide effective compensation for model uncertainty and disturbances, while learning-based methods can estimate unknown dynamics, disturbances, or uncertainty representations [1], [5], [13]–[18]. Networked-control research separately demonstrates that communication delay and packet loss can significantly influence closed-loop performance [3], [4], [22]–[25].

The present results provide an experimental indication that the combination of these concepts does not automatically produce an improvement across all operating conditions. In particular, the RBFNN provided a measurable benefit in the combined networked-uncertainty experiment but not in the experiments dominated by sensor degradation or physical dynamic uncertainty.

This distinction is relevant to the research objective because the proposed framework does not treat the learning model as a replacement for the underlying controller. Instead, the learning component provides an uncertainty estimate to an existing robust adaptive control mechanism. The results suggest that this architecture can be useful under some uncertainty configurations, but its benefit depends on the quality and relevance of the learned uncertainty representation.

Therefore, the experimental findings provide a more qualified interpretation of the proposed approach than a general claim of improved robustness. The contribution is better characterized as an investigation of when learning-based uncertainty estimation can and cannot provide additional value to robust adaptive control in a networked robotic system.

7.3 Robustness Analysis

The robustness analysis indicates that the baseline ASMC itself provides substantial tracking capability under several uncertainty conditions. In particular, the controller maintained bounded trajectories in the presence of communication delay, packet loss, sensor degradation, changing robot parameters, and external disturbances.

The combined-uncertainty experiment produced substantially larger tracking errors than the nominal and sensor-noise scenarios, demonstrating the cumulative effect of multiple uncertainty sources on the closed-loop system. The learning-enhanced controller reduced the average RMSE under the combined networked-uncertainty configuration, suggesting that the additional uncertainty estimate can partially compensate for information degradation and system uncertainty.

However, the robustness benefit was not universal. When sensor degradation was introduced, the learned uncertainty estimate did not compensate for the resulting measurement bias sufficiently to improve tracking. This highlights an important distinction between uncertainty associated with the plant dynamics and uncertainty associated with the measurement itself.

Consequently, robustness in the proposed architecture should not be interpreted as a single scalar property. Different uncertainty sources affect different parts of the feedback loop, and a learning model trained primarily on one class of uncertainty may not generalize effectively to another class.

7.4 Limitations

Several limitations should be considered when interpreting the results.

First, the experiments were conducted in simulation using a 2-DOF robotic manipulator model. The results therefore do not yet demonstrate performance on a physical robotic platform.

Second, the uncertainty models were deliberately defined within the simulation environment. The selected noise, bias, communication delay, packet-loss probability, parameter variations, and disturbances provide controlled experimental conditions but do not represent every possible real-world uncertainty profile.

Third, the RBFNN was trained using simulated data generated from the same general robotic system family used for evaluation. Although separate training and validation data were used, this does not establish generalization to substantially different robot dynamics or unseen hardware.

Fourth, the RBFNN architecture and controller gains were selected for the present simulation study. A different learning architecture, uncertainty representation, or controller design may produce different results.

Fifth, the multi-seed evaluation was performed for the main combined and sensor-degradation comparisons, whereas some other scenarios were evaluated using a single run. Therefore, those single-run results should be regarded as preliminary rather than statistically established conclusions.

Finally, processor-level execution time was not measured. The current computational analysis therefore addresses control effort but does not establish the real-time computational overhead of the RBFNN.

7.5 Practical Implications

The results indicate that learning-based uncertainty estimation should be integrated selectively within a control architecture rather than treated as a universal replacement for analytical robustness mechanisms.

For networked robotic systems, the approach may be useful when communication imperfections and system uncertainty interact in a way that can be represented from available state and network information. However, the results also show that measurement degradation may require a dedicated sensing, fault-detection, or state-estimation mechanism rather than relying solely on a learned lumped uncertainty estimate.

From an implementation perspective, retaining the robust adaptive controller as the primary control mechanism provides a fallback mechanism when the learned estimate is inaccurate. This separation also makes it possible to evaluate the learning component independently through ablation experiments.

The findings therefore support further investigation of uncertainty-specific learning architectures, particularly models that explicitly distinguish between plant uncertainty, measurement uncertainty, and communication-induced uncertainty. Such an approach may provide a more precise mechanism for determining which uncertainty should be compensated by learning and which should be handled by state estimation or robust control.

Conclusion and Future Work

This study investigated a learning-enhanced robust adaptive control framework for a networked two-degree-of-freedom robotic manipulator operating under multiple sources of uncertainty. The framework combined an adaptive sliding-mode controller with an RBF neural-network-based uncertainty estimation component. Sensor noise, sensor degradation, communication delay, packet loss, uncertain robot dynamics, and external disturbances were considered in the experimental evaluation.

The results demonstrated that the effect of the learning-based uncertainty estimator depended on the uncertainty configuration. Under the combined networked-uncertainty scenario, the ASMC + RBFNN configuration achieved lower mean tracking RMSE than the ASMC baseline across ten random seeds, with reductions of 5.35% for the first joint and 18.17% for the second joint. However, the learning component did not provide an improvement when sensor degradation was additionally introduced, and the changing-dynamics and external-disturbance experiment likewise did not demonstrate a reduction in tracking RMSE. These

findings indicate that learning-based uncertainty estimation cannot be assumed to improve control performance uniformly across different uncertainty sources.

The study therefore provides evidence for a more conditional interpretation of learning-enhanced robust adaptive control. The effectiveness of the learning component depends on the type of uncertainty represented in the training data, the relationship between the estimated uncertainty and the control input, and the manner in which the estimate is incorporated into the control law.

The current study is limited to simulation-based evaluation of a 2-DOF robotic manipulator. Future work should investigate the framework on a physical robotic platform and examine uncertainty models derived from experimentally collected sensor and communication data. Additional work should also investigate uncertainty estimators that explicitly distinguish between plant, sensor, and communication uncertainty rather than representing them through a single lumped estimate. Further studies could evaluate alternative learning architectures, different robust-adaptive controllers, broader communication conditions, and real-time computational requirements.

The final conclusions should be interpreted within the uncertainty ranges, robotic model, controller configuration, and simulation conditions investigated in this study.

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